EXPERIMENT - 7

AIM:

To write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. simulate the flow of frames from one node to another.

CODE

SENDER.PY

```
import time
w = int (input ("Enter window size: "))
msg = input ("Enter Message: ")
frames = [[i, ch] for i, ch in enumerate (msg)]
base = 0
while base < len (frames):
    with open ("sender_buffer.txt", "w") as f:
        for i in range (base, min (base+w, len (frames))):
            f.write (f"{frames[i][0]},
                {frames[i][1]} \n")
    Print ("\n Frames Sent:")
    for i in range (base, (min (base+w) len (frames))):
        Print (frames[i])
    time.sleep (3)

my:
    with open ("Receiver_buffer.txt" "r") as f:
        data = f.read().strip()
    if data.startswith ("ACK"):
        ack = int (data.split (":") [1])
    if ack == base + w - 1 and ack >= len (frames) - 1:
        Print ("ACK received:", ack)
        base = ack + 1
else:
    Print ("wrong ACK Resending...")
```

Scanned with OKEN Scanner

```
elif data.startswith("NACK"):
    print("NACK Received. Resending ....")

except:
    print("waiting for Receiver ....")
```

RECEIVER.PY

```
exp = 0
with open("sender_Buffer.txt", "r") as f:
    lines = f.readlines()

ok = True
for line in lines:
    no, data = line.strip().split(",")
    if int(no) != exp:
        ok = False
        nack = exp
        break
    exp += 1
with open("Receiver_Buffer.txt", "w") as f:
    if ok:
        f.write(f"ACK:{exp-1}")
        print("ACK sent:", exp-1)
    else:
        f.write(f"NACK:{nack}")
        print("NACK sent:", nack)
```

INPUT:

```
Enter window size: 3
Enter Message: HELLO
```

OUTPUT (sender)

```
Frames Sent:
[0, 'H']
[1, 'E']
[2, 'L']
Ack received: 2
```

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Frames sent :

[3, 'L']

[4, 'O']

Ack received : 4

OUTPUT (Receiver)

ACK sent : 2

ACK sent : 4

RESULT :

Thus the Program to implement flow control at data link layer used SLIDING WINDOW has been executed successfully